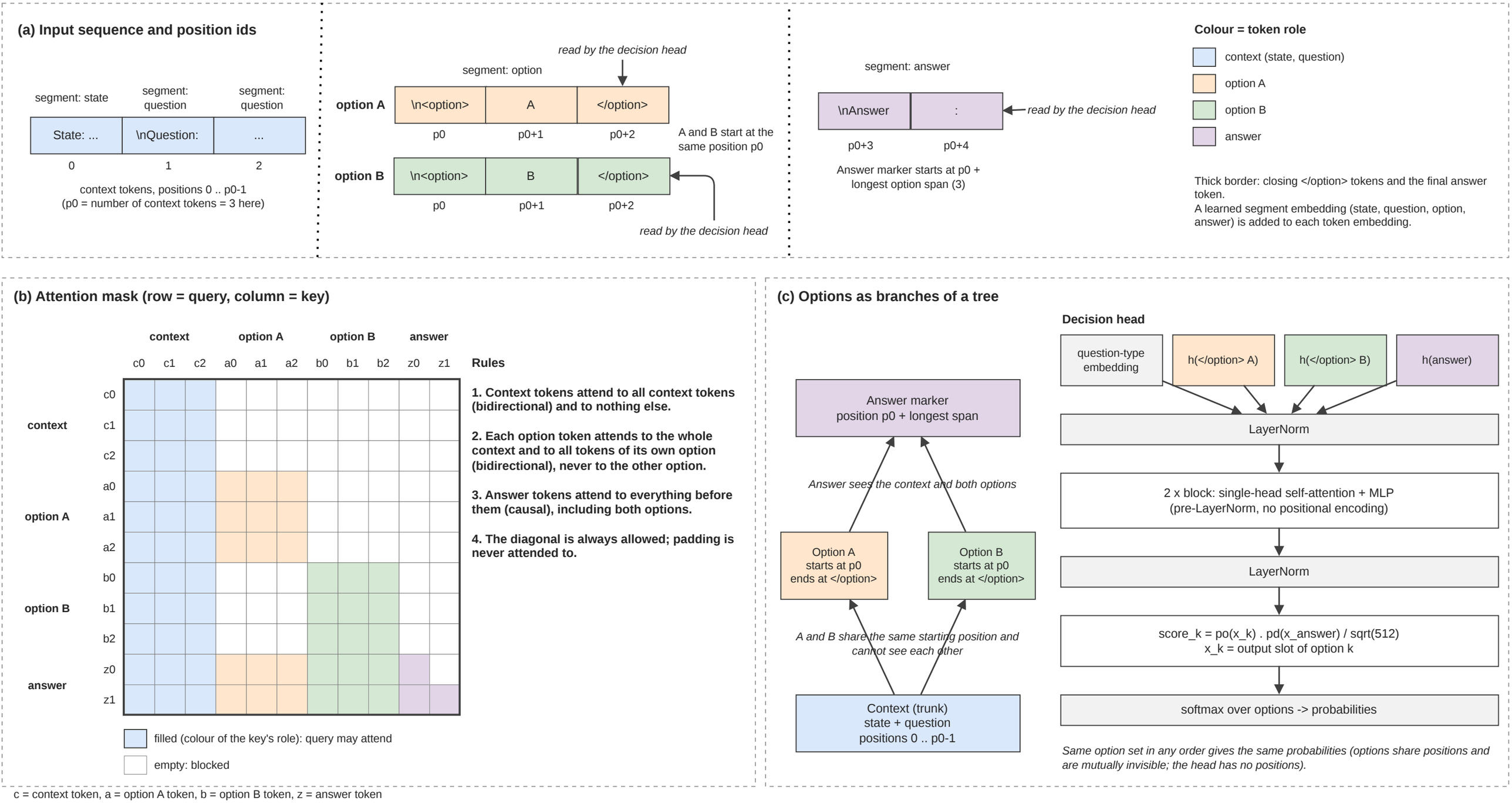

(a) Input sequence and position ids
segment: state
segment: question
segment: question
State: ...
\nQuestion:
...
0
1
2
context tokens, positions 0 .. p0-1
(p0 = number of context tokens = 3 here)
read by the decision head
segment: option
option A
\n<option>
A
</option>
p0
p0+1
p0+2
A and B start at the same position p0
option B
\n<option>
B
</option>
p0
p0+1
p0+2
read by the decision head
segment: answer
\nAnswer
:
read by the decision head
p0+3
p0+4
Answer marker starts at p0 + longest option span (3)
Colour = token role
context (state, question)
option A
option B
answer
Thick border: closing </option> tokens and the final answer token.
A learned segment embedding (state, question, option, answer) is added to each token embedding.
(b) Attention mask (row = query, column = key)
context
option A
option B
answer
c0 c1 c2 a0 a1 a2 b0 b1 b2 z0 z1
filled (colour of the key's role): query may attend
empty: blocked
Rules
1. Context tokens attend to all context tokens (bidirectional) and to nothing else.
2. Each option token attends to the whole context and to all tokens of its own option (bidirectional), never to the other option.
3. Answer tokens attend to everything before them (causal), including both options.
4. The diagonal is always allowed; padding is never attended to.
c = context token, a = option A token, b = option B token, z = answer token
(c) Options as branches of a tree
Answer marker
position p0 + longest span
Answer sees the context and both options
Option A
starts at p0
ends at </option>
Option B
starts at p0
ends at </option>
A and B share the same starting position and cannot see each other
Context (trunk)
state + question
positions 0 .. p0-1
Decision head
question-type embedding
h(</option> A)
h(</option> B)
h(answer)
LayerNorm
2 x block: single-head self-attention + MLP
(pre-LayerNorm, no positional encoding)
LayerNorm
score_k = po(x_k) . pd(x_answer) / sqrt(512)
x_k = output slot of option k
softmax over options -> probabilities
Same option set in any order gives the same probabilities (options share positions and are mutually invisible; the head has no positions).